

DEEPFAKE IMAGE DETECTION USING DEEP LEARNING METHODS

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The Problem

Detecting Synthetic Faces

- GAN-generated images visually indistinguishable from real photographs
- Subtle artifacts present: frequency anomalies, texture inconsistencies, boundary blending
- These signals are invisible to the human eye but learnable by a classifier
- Key challenge: do models learn general fakeness, or generation-method-specific patterns?

Dataset & Experimental Design

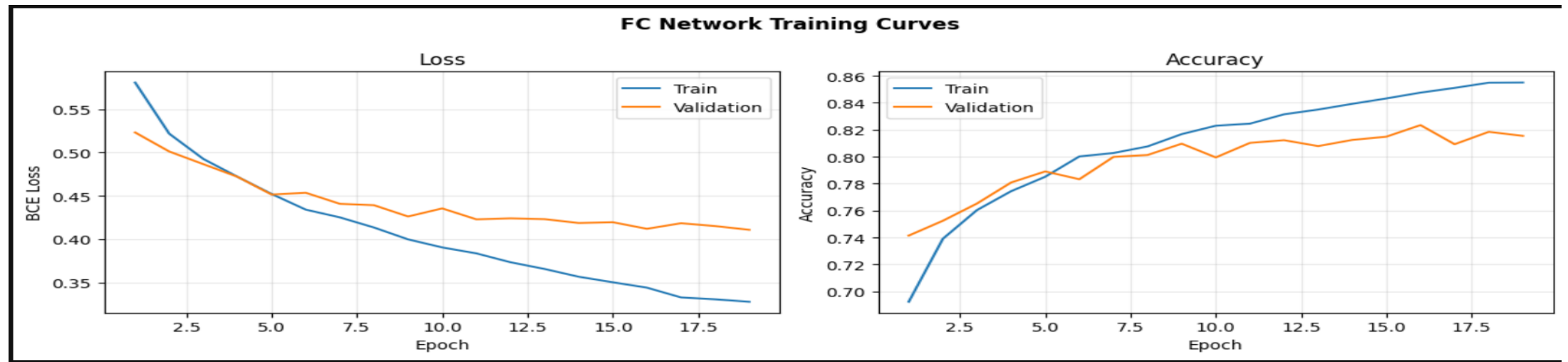
Dataset and Approach

- 40.000 total images: 30k train / 5k validation / 5k test
- Perfectly balanced: 50% real, 50% fake in every split
- Test set withheld until all models finalized and evaluated exactly once
- Additional out-of-distribution set: 3,000 Stable Diffusion images (all fake)
- One variable changed at a time across all three stages

Stage 1: FC Baseline

Hand-Crafted Features + FC Network

- Four features extracted per image: LBP, FFT, DCT, Wavelet
- Combined into a 201-dimensional vector per image
- MLP architecture: 201 -> 512 -> 128 -> 1 (170K parameters)
- Validation accuracy: **81.54%** / AUC: **0.90**
- Limitation: features discard all spatial information



Stage 2: CNN Architecture Design

Stage 2: Three Custom CNN Architectures

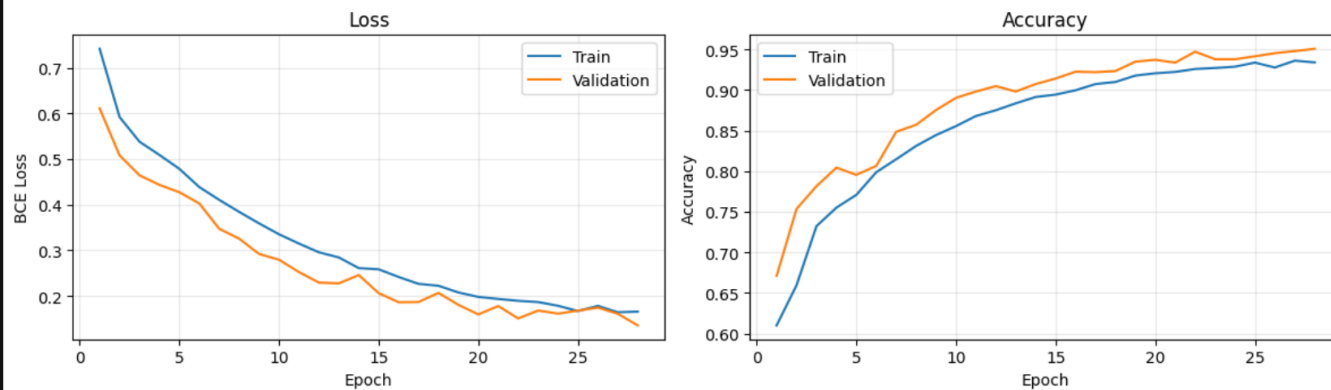
1. CNN-A: 3 conv blocks + flatten head, 8.48M parameters
 - 1.1 Large FC layer memorises spatial positions
2. CNN-B: 4 conv blocks + Global Average Pooling, 390K parameters
 - 2.1 GAP averages each feature map globally; forces spatially invariant features
3. CNN-C: identical to CNN-B + training augmentation
 - 3.1 Flip, colour jitter, small rotation

Stage 2: Results & Key Findings

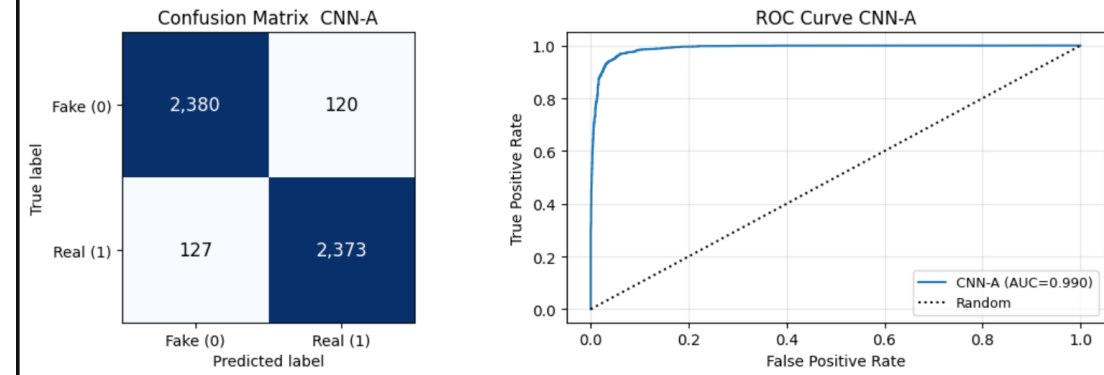
Model	Accuracy (Val)	ROC-AUC (Val)	Parameters
CNN-A	95.06%	0.99	8,482,817
CNN-B	96.18%	0.99	389,633
CNN-C	93.76%	0.98	389,633
FC Baseline (Stage 1, ref.)	81.54%	0.90	170,497

- All CNNs: +13 to 15 percentage points over Stage 1
- CNN-B outperforms CNN-A with 22x fewer parameters
- Augmentation hurt: texture-level signals eroded by colour and geometric transforms
- Architecture design matters more than parameter count

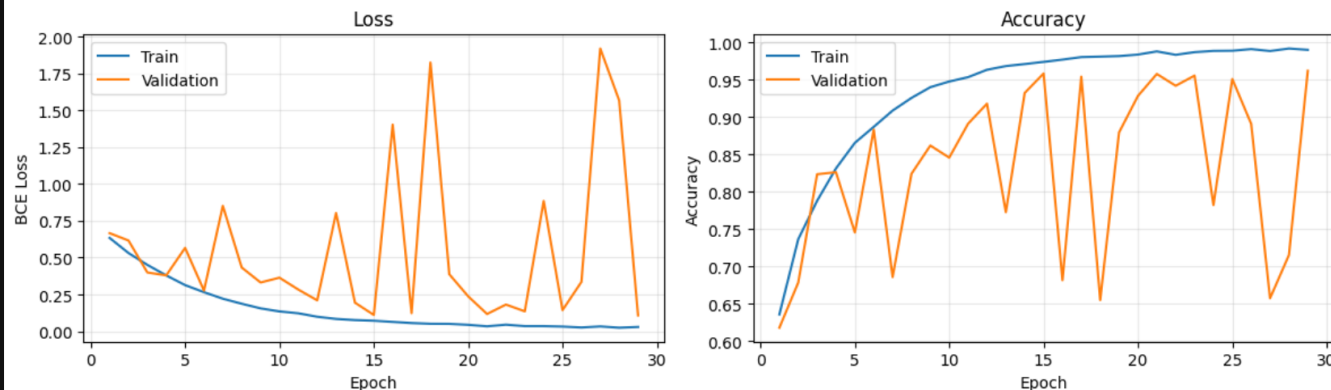
CNN-A Training Curves



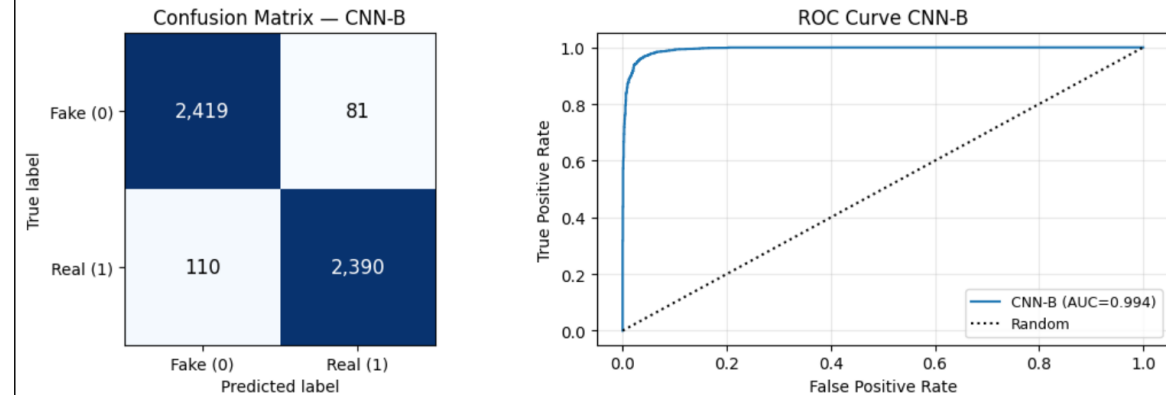
CNN-A Evaluation



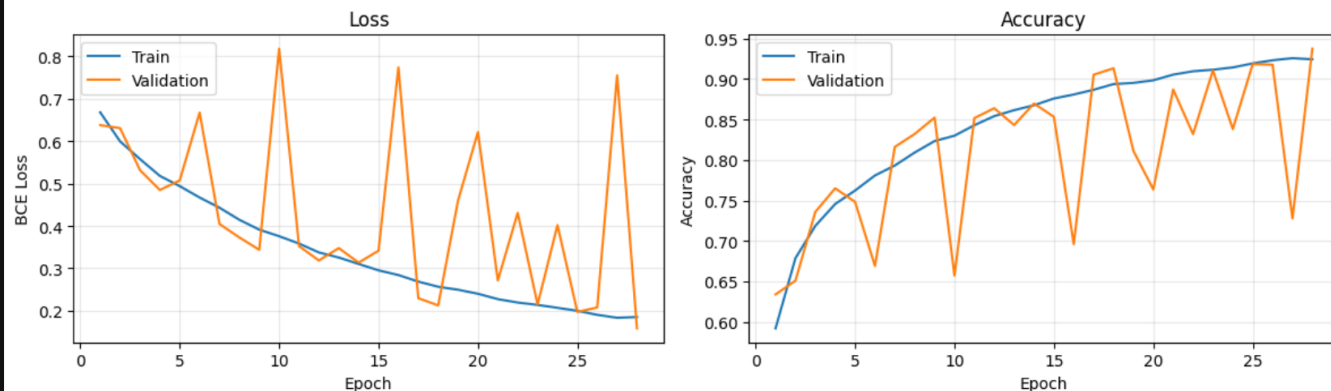
CNN-B Training Curves



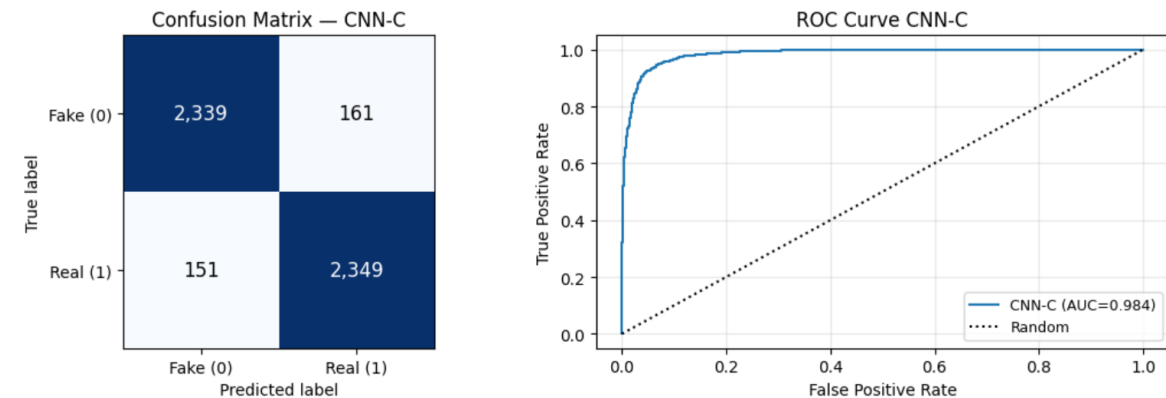
CNN-B Evaluation



CNN-C Training Curves



CNN-C Evaluation



Stage 3: Pretrained Models

Transfer Learning

- **ResNet-18:** 11.2M parameters, residual connections, full fine-tuning
- **EfficientNet-B0:** 4.01M parameters, compound scaling, full fine-tuning
- Learning rate: 0.001
- ReduceLROnPlateau scheduler: LR halved after 3 epochs without improvement

Model	Accuracy (Val)	ROC-AUC (Val)	Parameters
ResNet-18	95.56%	0.9895	11,177,025
EfficientNet-B0	96.90%	0.9958	4,008,829
CNN-B (Stage 2, ref.)	96.18%	0.9939	389,633

Final Test Results

- Validation-to-test gaps uniformly small: methodology was sound
- EfficientNet-B0: best result across all stages

Model	Accuracy	F1-Score	ROC-AUC	Parameters
FC Baseline (Stage 1)	81.64%	81.55%	0.8945	170,497
CNN-A (Stage 2)	94.98%	94.96%	0.9898	8,482,817
CNN-B (Stage 2)	95.94%	95.93%	0.9921	389,633
CNN-C (Stage 2)	93.14%	93.17%	0.9826	389,633
ResNet-18 (Stage 3)	95.24%	95.23%	0.9891	11,177,025
EfficientNet-B0 (Stage 3)	97.22%	97.20%	0.9969	4,008,829

Cross-Distribution Test: Stable Diffusion

- 3,000 Stable Diffusion images all fake
- Stable Diffusion: diffusion-based process, entirely different artifact profile

Model	Detected as Fake	Detection Rate
FC Baseline	25 / 3,000	0.8%
CNN-A	3 / 3,000	0.1%
CNN-B	40 / 3,000	1.3%
CNN-C	50 / 3,000	1.7%
ResNet-18	2 / 3,000	0.1%
EfficientNet-B0	54 / 3,000	1.8%

Conclusions

Key Takeaways:

- Spatial learning (CNN) beats hand-crafted features
- Architecture design outweighs parameter count: CNN-B proves this
- Pretrained models improve results, but do not automatically dominate

Future work:

- Train on diverse generation methods: GANs, diffusion models, VAEs
- Frequency-domain augmentation to simulate cross-distribution variation